Microeconometrics Using Stata

LINEAR INSTRUMENTAL - VARIABLES REGRESSION

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Exercise 1

Exercise 1

Exercise 1

Estimate by **2SLS** the same regression model as in **section 7.4.4**, with the instruments **multlc** and **firmsz**. Compare the 2SLS estimates with OLS estimates. Perform a test of endogeneity of **hi_empunion**. Perform a test of overidentification. State what you conclude. Throughout this exercise, perform inference that is robust to heteroskedasticity.

Exercise 2

Exercise 2

Exercise 2

Repeat exercise 1 using optimal **GMM**.

Exercise 3

Exercise 3

Exercise 3

Use the model and instruments of exercise 1. Compare the following estimators: **2SLS**, **LIML**, and optimal **GMM** given heteroskedastic errors. For the last model, estimate the parameters by using the community-contributed **ivreg2** command in addition to **ivregress**.

Exercise 4

Exercise 4

Exercise 4

Use the model of exercise 1. Compare **2SLS** estimates as the instruments **ssiratio**, **lowincome**, **multlc**, and **firmsz** are progressively added.

Exercise 5

Exercise 5

Exercise 5

Use the model and instruments of exercise 1. Perform appropriate diagnostics and tests for weak instruments using the **2SLS** estimator. State what you conclude. Throughout this exercise, perform inference assuming errors are i.i.d.

Exercise 6

Exercise 6

Exercise 6

Use the model and instruments of exercise 1. Use the community-contributed **condivreg** command to perform inference for the **2SLS** estimator. Compare the results with those using conventional asymptotics.

Exercise 7

Exercise 7

Exercise 7

Use the model and instruments of exercise 1. Use the community-contributed **jive** command, and compare estimates and standard errors from the four different variants of **JIVE** and from optimal **GMM**. Throughout this exercise, perform inference that is robust to heteroskedasticity.

Exercise 8

Exercise 8

Exercise 8

Fit the **3SLS** model of **section 7.10**, and compare the **3SLS** coefficient estimates and standard errors in the **ldrugexp** equation with those from **2SLS** estimation (with default standard errors).

Exercise 9

Exercise 9

This question considers the same earnings-schooling dataset as that analyzed in **Cameron and Trivedi** (**2005**, 111). The data are in **mus207klingdata.dta**. The **describe** command provides descriptions of the regressors. There are three endogenous regressors (years of schooling, years of work experience, and experience-squared) and three instruments (a college proximity indicator, age, and age-squared). Interest lies in the coefficient of schooling. Perform appropriate diagnostics and tests for weak instruments for the following model. State what you conclude. The following commands yield the **IV** estimator: see commands

Exercise 10

Exercise 10

Exercise 10

Use the same dataset as the previous question. Treat only **grade76** as endogenous, let **exp76** and **expsq76** be exogenous, and use **col4** as the only instrument. Perform appropriate diagnostics and tests for a weak instrument, and state what you conclude. Then, use the community-contributed **condivreg** command to perform inference, and compare the results with those using conventional asymptotics.

Exercise 11

Exercise 11

When an endogenous variable enters the regression nonlinearly, the obvious **IV** estimator is inconsistent and a modification is needed. Specifically, suppose $y_1 = \beta y_2^2 + u$ and the first-stage equation for y_2 is $y_2 = \pi_2 z + v$ where the zero-mean errors and are correlated. Here the endogenous regressor appears in the structural equation as y_2^2 rather than y_2. The **IV** estimator is

$$\hat{\beta}_{IV} = \left(\sum_i z_i y_{2i}^2 \right)^{-1} \sum_i z_i y_{1i} \tag{1}$$

This can be implemented by a regular **IV** regression of y on y_2^2 with the instrument z: regress y_2^2 on z and then regress y_1 on the first-stage prediction $\widehat{y_2^2}$. If instead we regress y_2 on z at the first stage, giving $\hat{y_2}$, and then regress y_1 on $(\hat{y}_2)^2$, an inconsistent estimate is obtained. Generate a simulation sample to demonstrate these points. Consider whether this example can be generalized to other nonlinear models where the nonlinearity is in regressors only, so that

$$y_1 = \mathbf{g}(\mathbf{y}_2)' \beta + u \tag{2}$$

where $\mathbf{g}(\mathbf{y}_2)$ is a nonlinear function of $\mathbf{y}_2$.

Exercise 12

Exercise 12

Exercise 12

This exercise is based on artificially generated data. Set sample size to $1,000$. Set seed to **10101**. Generate a sample using the following **DGP**: $u \sim N(0,1)$; $e \sim 0.6u + N(0,1)$; $z \sim uniform(0,1)$; $y_2 = z + 0.5z^2 + 0.0625z^3 + e$; $x \sim N(0,9)$; $y_1 = y_2 + 0.5y_2^2 + x + u$ Estimate the y_1 equation by **OLS**. Theoretically, the estimates of the equation are inconsistent. Compare all the slope coefficients with their true values, and discuss whether the direction of the bias is as you might expect. **OLS** is inconsistent in this case, yet some coefficients are estimated close to their true values; explain why. Run the same regression again with the intercept constrained to zero, and comment on any differences that you might observe.

Exercise 13

Exercise 13

Exercise 13

Continuing the previous exercise, estimate three variants of the y_1-equation by **2SLS** using as instruments

i) x, z, and z^2

ii) x, z, z^2, and z^3

iii) $\hat{y}_2$ and $\hat{y}_2^2$

where $\hat{y}_2$ is the fitted value of y_2 from the reduced form of y_2. Are the instruments chosen in each case valid and relevant? Which of the three estimates would you prefer on theoretical grounds? Explain your answer. With reference to the **2SLS** estimates used earlier, apply the test of overidentification where appropriate, and interpret your results.

Exercise 14

Exercise 14

Exercise 14

This question continues with the model presented in question 12 above. Modify the specification of the y_1 equation such that the equation errors are heteroskedastic, with skedasticity being a quadratic function of x; that is,

$$Var(u) = h(x) \times N(0,1) \tag{3}$$

Reestimate the y_1 equation by **2SLS** and two-step **GMM** using as instruments x, z, z^2, z^3. In both cases, apply the test of overidentifying restrictions. Compare the results, and comment on the outcome of the test.

References I

Cameron, A. C. and Trivedi, P. K. (2022a).
Microeconometrics Using Stata: Volume I: Cross-Sectional and Panel Regression Methods.
Stata Press.

Cameron, A. C. and Trivedi, P. K. (2022b).
Microeconometrics Using Stata: Volume II: Nonlinear Models and Causal Inference Methods.
Stata Press.

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Commands

```
. use mus207klingdata.dta, clear

. global x2list black south76 smsa76 reg2-reg9 smsa66
> sinmom14 nodaded nomomed daded momed famed1-famed8

. ivregress 2sls wage76 (grade76 exp76 expsq76 = col4 age76 agesq76) $x2list,
> vce(robust) perfect

. estat firststage
```

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